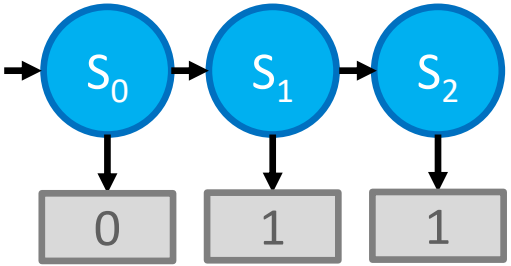
BKT - Inference

Machine Learning for Behavioral Data

March 10, 2026

Inference Example



$$p_0 = 0.5$$
$$p_S = 0.2$$
$$p_G = 0.3$$
$$p_L = 0.4$$
$$p_F = 0.0$$

S_0	$p(S_0)$
1	p_0
0	$1\text{-}p_0$

S_t	S_{t+1}	$p(S_{t+1}\|S_t)$
0	0	$1 - p_L$
0	1	p_L
1	0	p_F
1	1	$1 - p_F$

S_t	O_t	$p(O_t\|S_t)$
0	0	$1 - p_G$
0	1	p_G
1	0	p_S
1	1	$1 - p_S$

Some useful rules:

$$p(A, B) = p(A|B) \cdot p(B)$$

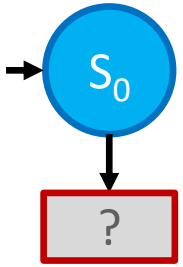
$$p(A|B) = \frac{p(B|A) \cdot p(A)}{p(B)}$$

$$p(A = 1) = p(A = 1, B = 1) + p(A = 1, B = 0)$$

- $p(s_0 = 1)$?
- $p(o_0 = 1)$?
- $p(s_1 = 1|o_0 = 0)$?
- $p(o_1 = 1|o_0 = 0)$?
- $p(s_2 = 1|o_0 = 0, o_1 = 1)$?

Inference in BKT models

Equations for time step 0:



$$p(s_0 = 1) = p_0$$

$$p(s_0 = 0) = 1 - p_0$$

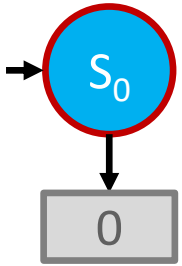
$$p(o_0 = 1) = p(o_0 = 1, s_0 = 1) + p(o_0 = 1, s_0 = 0)$$
$$= (1 - p_s) \cdot p_0 + p_G \cdot (1 - p_0)$$

$$p(o_0 = 0) = p_S \cdot p_0 + (1 - p_G) \cdot (1 - p_0)$$

Inference in BKT models

$$\underbrace{p(s_0 = 1 | o_0 = 0)}_{p_{s_0|0}} = \frac{p(o_0 = 0 | s_0 = 1) \cdot p(s_0 = 1)}{p(o_0 = 0)}$$

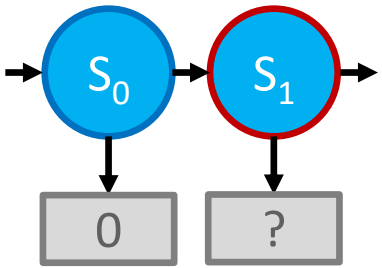
$$= \frac{p_S \cdot p_0}{p_s \cdot p_0 + (1 - p_G) \cdot (1 - p_0)}$$



$$p(s_0 = 0 | o_0 = 0) = 1 - p_{s_0|0}$$

Inference in BKT models

Equations for time step 1:



$$p(s_1 = 1|o_0 = 0) = \frac{p(s_1 = 1, o_0 = 0)}{p(o_0 = 0)}$$

$$= \frac{p(s_1 = 1, s_0 = 1, o_0 = 0)}{p(o_0 = 0)} + \frac{p(s_1 = 1, s_0 = 0, o_0 = 0)}{p(o_0 = 0)}$$

$$= \frac{p(s_1 = 1|s_0 = 1) \cdot p(o_0 = 0|s_0 = 1) \cdot p(s_0 = 1)}{p(o_0 = 0)} + \frac{p(s_1 = 1|s_0 = 0) \cdot p(o_0 = 0|s_0 = 0) \cdot p(s_0 = 0)}{p(o_0 = 0)}$$

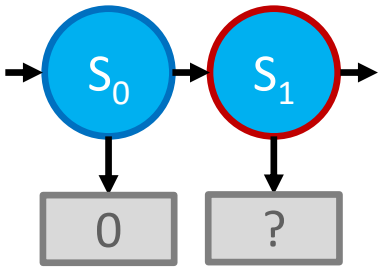
$$p(s_1 = 1|o_0 = 0) = (1 - p_F) \cdot p_{s_0|0} + p_L \cdot (1 - p_{s_0|0})$$

Inference in BKT models

$$p(s_1 = 1|o_0 = 1) = (1 - p_F) \cdot p_{s_0|1} + p_L \cdot (1 - p_{s_0|1})$$

$$p(s_1 = 1|o_0 = 0) = (1 - p_F) \cdot p_{s_0|0} + p_L \cdot (1 - p_{s_0|0})$$

$$p_{s_1|o_0} = (1 - p_F) \cdot p_{s_0|o_0} + p_L \cdot (1 - p_{s_0|o_0})$$



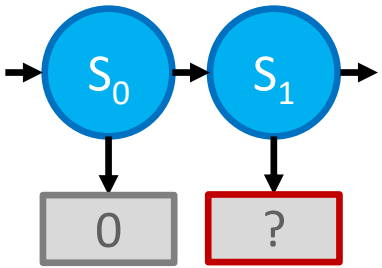
Inference in BKT models

$$p(o_1 = 1|o_0 = 0) = \frac{p(o_1 = 1, o_0 = 0)}{p(o_0 = 0)}$$

$$+ \frac{p(o_1 = 1, s_1 = 1, s_0 = 1, o_0 = 0)}{p(o_0 = 0)}$$

$$+ \frac{p(o_1 = 1, s_1 = 1, s_0 = 0, o_0 = 0)}{p(o_0 = 0)}$$

$$+ \frac{p(o_1 = 1, s_1 = 0, s_0 = 1, o_0 = 0)}{p(o_0 = 0)} + \frac{p(o_1 = 1, s_1 = 0, s_0 = 0, o_0 = 0)}{p(o_0 = 0)}$$



$$= p(o_1 = 1|s_1 = 1) \cdot (p(s_1 = 1|s_0 = 1) \cdot p(o_0 = 0|s_0 = 1) \cdot p(s_0 = 1)/p(o_0 = 0))$$
$$+ p(o_1 = 1|s_1 = 1) \cdot (p(s_1 = 1|s_0 = 0) \cdot p(o_0 = 0|s_0 = 0) \cdot p(s_0 = 0)/p(o_0 = 0))$$
$$+ p(o_1 = 1|s_1 = 0) \cdot (p(s_1 = 0|s_0 = 1) \cdot p(o_0 = 0|s_0 = 1) \cdot p(s_0 = 1)/p(o_0 = 0))$$
$$+ p(o_1 = 1|s_1 = 0) \cdot (p(s_1 = 0|s_0 = 0) \cdot p(o_0 = 0|s_0 = 0) \cdot p(s_0 = 0)/p(o_0 = 0))$$

$$p(o_1 = 1|o_0 = 0) = (1 - p_S) \cdot p_{s_1|o_0=0} + p_G \cdot (1 - p_{s_1|o_0=0})$$

Inference in BKT models

$\boldsymbol{o_{t-1}} = [o_0, \dots, o_{t-1}]$

Equations for time $t = 0$:

Belief about latent state before observation

$$p(s_0 = 1) = p_0$$

Predicted observation at time t

$$p(o_0 = 1) = (1 - p_S) \cdot p_0 + p_G \cdot (1 - p_0)$$

$$p(o_0 = 0) = p_S \cdot p_0 + (1 - p_G) \cdot (1 - p_0)$$

Posterior: belief about latent state after observation

$$\left.\begin{aligned} p_{s_0|1} &= \frac{(1 - p_S) \cdot p_0}{(1 - p_S) \cdot p_0 + p_G \cdot (1 - p_0)} \\ p_{s_0|0} &= \frac{p_S \cdot p_0}{p_S \cdot p_0 + (1 - p_G) \cdot (1 - p_0)} \end{aligned}\right\} p_{s_0|o_0}$$

Equations for time steps $t = 1, \dots, T$:

$$p_{s_t|\boldsymbol{o_{t-1}}} = (1 - p_F) \cdot p_{s_{t-1}|\boldsymbol{o_{t-1}}} + p_L \cdot (1 - p_{s_{t-1}|\boldsymbol{o_{t-1}}})$$

$$p(o_t = 1|\boldsymbol{o_{t-1}}) = (1 - p_S) \cdot p_{s_t|\boldsymbol{o_{t-1}}} + p_G \cdot (1 - p_{s_t|\boldsymbol{o_{t-1}}})$$

$$p(o_t = 0|\boldsymbol{o_{t-1}}) = p_S \cdot p_{s_t|\boldsymbol{o_{t-1}}} + (1 - p_G) \cdot (1 - p_{s_t|\boldsymbol{o_{t-1}}})$$

$$\left.\begin{aligned} p_{s_t|1,\boldsymbol{o_{t-1}}} &= \frac{(1 - p_S) \cdot p_{s_t|\boldsymbol{o_{t-1}}}}{(1 - p_S) \cdot p_{s_t|\boldsymbol{o_{t-1}}} + p_G \cdot (1 - p_{s_t|\boldsymbol{o_{t-1}}})} \\ p_{s_t|0,\boldsymbol{o_{t-1}}} &= \frac{p_S \cdot p_{s_t|\boldsymbol{o_{t-1}}}}{p_S \cdot p_{s_t|\boldsymbol{o_{t-1}}} + (1 - p_G) \cdot (1 - p_{s_t|\boldsymbol{o_{t-1}}})} \end{aligned}\right\} p_{s_t|\boldsymbol{o_t}}$$

[Corbett & Anderson, UMUAI 1994]